



SS-NMDE: Boosting Night-time Monocular Depth Estimation Using Semantic Informations

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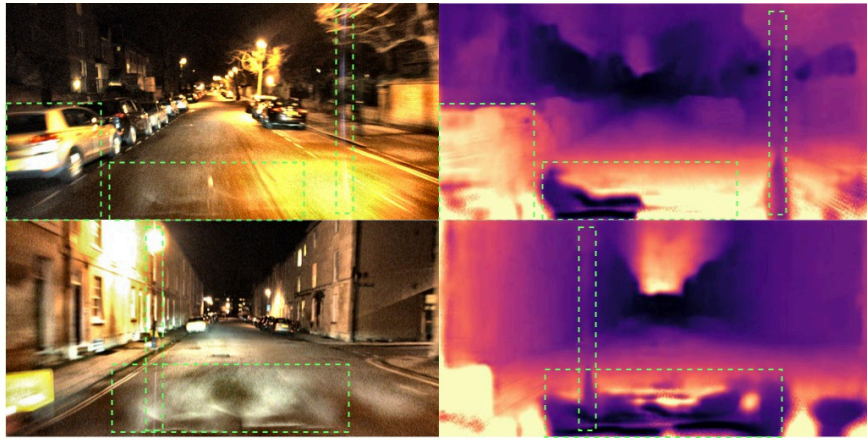
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Introduction

We proposed a **night-time monocular depth estimation network** which combine a *paired synthetic images training strategy* and a *depth estimation features boosting with semantic information from multi-tasking training framework* with semantic segmentation problem. Our model can also extend to perform well on both day and night. Experiments show **SOTA results** on Oxford RobotCar and comparable results on nuScenes.

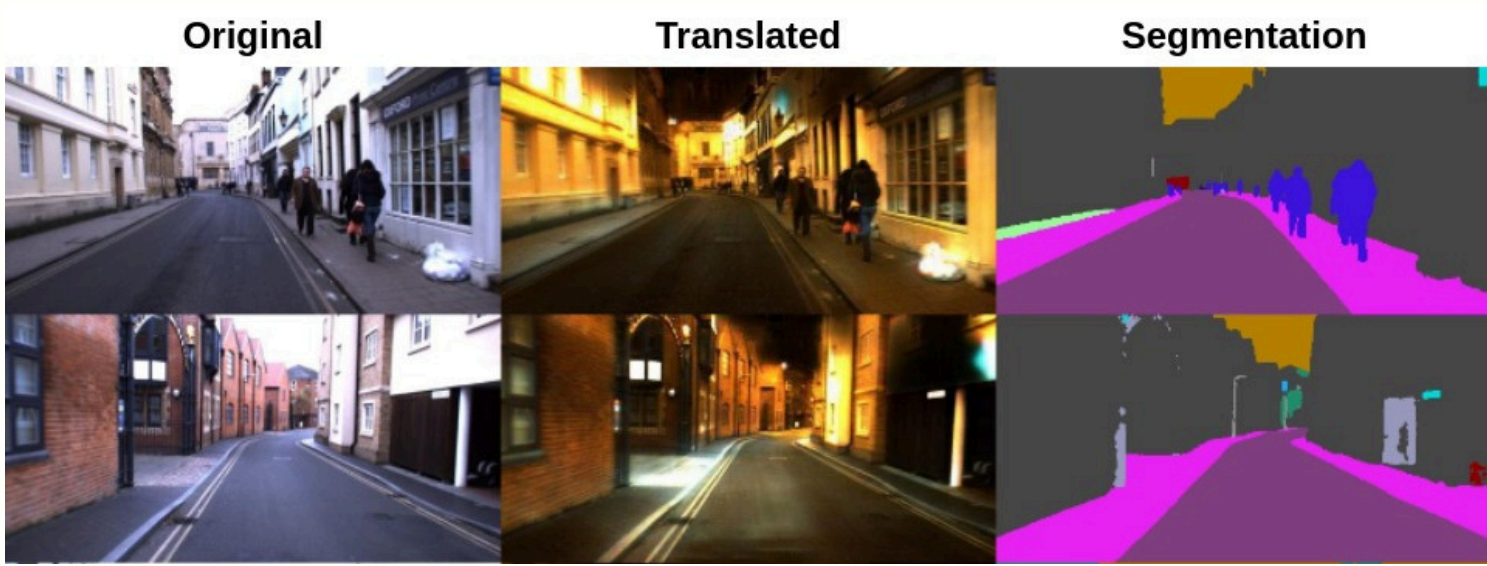
Problems & Motivation

Existing monocular depth estimation for day domain cannot perform well when applying on adverse domains like in day-time.



Monodepth2 problems on night-time images

Motivated by the synthetic paired training framework in DarkMDE. We found it lack semantic information for inferring consistency depth map. Therefore, we utilized the multi-tasking strategy for boosting the MDE feature map.



The translated day-to-night images and semantic segmentation informations

Experiments & Results

Achieved SOTA on Oxford RobotCar dataset and comparable results on nuScenes dataset with a full self-supervised learning framework (md4all had used supervised information for camera pose estimation):

- **SS-NMDE-N** for model trained on only night-time domain.
- **SS-NMDE-DN** for model trained on both night-time and day-time domain.

Method	Size	Domain	abs_rel	sq_rel	rmse	rmse_log	a1	a2	a3
md4all Split Night - 50m									
Monodepth2	320x544	d + n	0.211	2.853	4.705	0.280	0.823	0.938	0.964
DeFeat-Net	320x544	d + n	0.184	1.908	5.410	0.254	0.783	0.913	0.961
ADDS-Depth	320x544	d + n	0.197	1.027	4.111	0.256	0.680	0.924	0.970
WTSGD	320x544	d + n	0.174	1.637	6.302	0.245	0.754	0.915	0.964
STEPS	320x544	n	0.197	1.887	5.240	0.252	0.734	0.923	0.968
md4all	320x544	d + n	0.122	0.784	3.604	0.196	0.849	0.951	0.978
SS-NMDE-DN	320x544	d + n	0.120	0.684	3.368	0.189	0.849	0.957	0.983
md4all Split Day - 50m									
Monodepth2	320x544	d + n	0.122	0.782	3.408	0.182	0.864	0.956	0.979
DeFeat-Net	320x544	d + n	0.164	1.629	4.808	0.230	0.812	0.928	0.967
ADDS-Depth	320x544	d + n	0.131	0.692	3.121	0.188	0.847	0.951	0.980
WTSGD	320x544	d + n	0.176	1.603	6.036	0.245	0.750	0.912	0.963
STEPS	320x544	n	0.173	0.781	3.380	0.217	0.716	0.944	0.986
md4all	320x544	d + n	0.113	0.648	3.206	0.186	0.871	0.954	0.977
SS-NMDE-DN	320x544	d + n	0.116	0.653	3.103	0.174	0.870	0.959	0.983

Method	Size	Domain	abs_rel	sq_rel	rmse	rmse_log	a1	a2	a3
nuScenes night - 80m									
Monodepth2	320x576	d + n	1.185	42.306	21.613	1.567	0.184	0.36	0.504
RNW	320x576	d + n	0.326	3.999	9.932	0.417	0.492	0.765	0.870
STEPS	320x576	n	0.287	3.363	9.120	0.390	0.572	0.805	0.908
md4all	320x576	d + n	0.192	2.369	8.497	0.328	0.710	0.863	0.933
SS-NMDE-DN	320x576	d + n	0.212	2.161	8.784	0.342	0.657	0.849	0.928
nuScenes day - 80m									
Monodepth2	320x576	d	0.138	1.561	6.619	0.231	0.832	0.941	0.974
RNW	320x576	d + n	0.287		9.185		0.562		
STEPS	320x576	n	0.258	2.823	9.864	0.369	0.580	0.815	0.908
md4all	320x576	d + n	0.137	1.751	6.452	0.231	0.846	0.942	0.971
SS-NMDE-DN	320x576	d + n	0.145	1.778	6.689	0.232	0.827	0.940	0.974

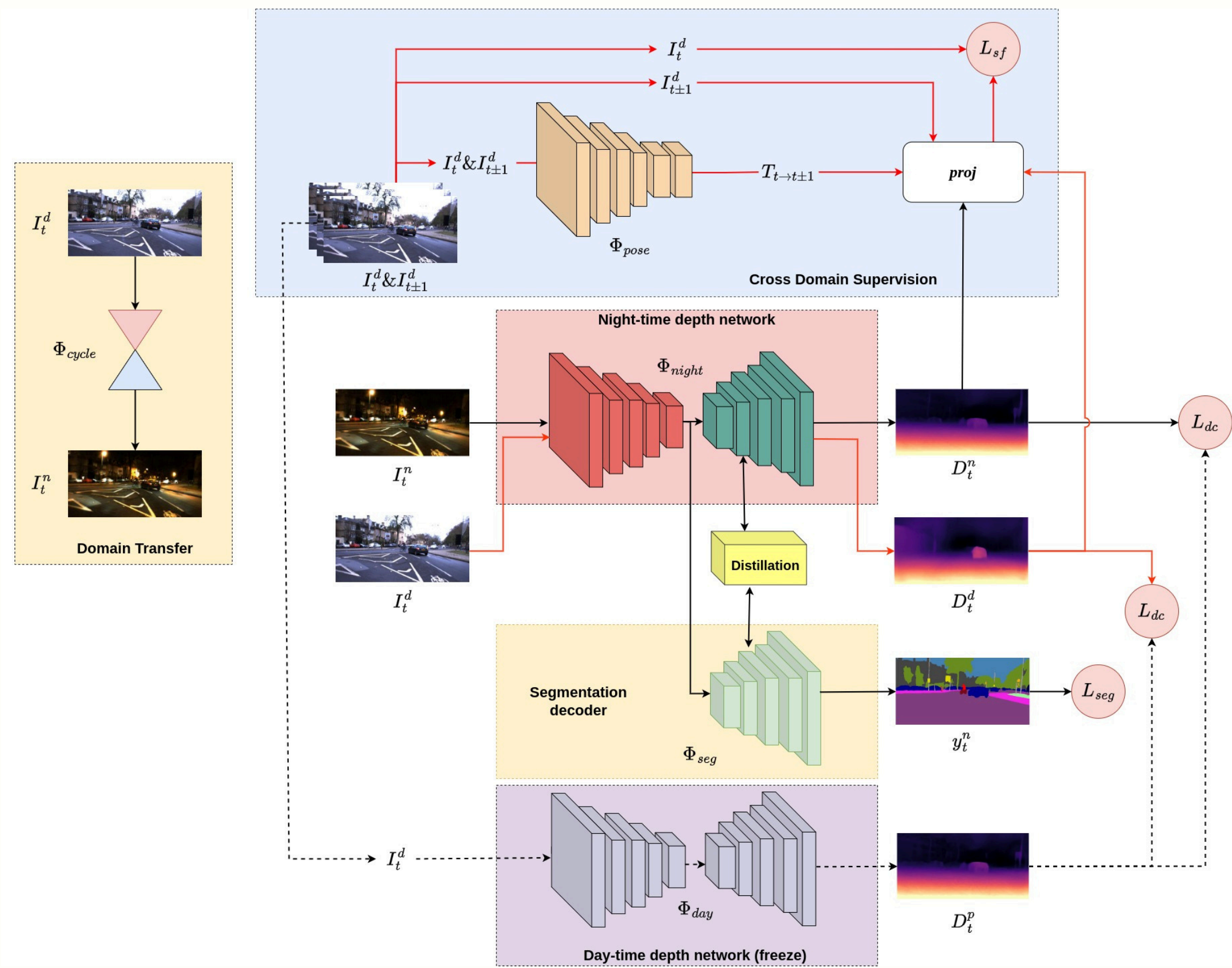
Experiments results on **Oxford RobotCar** and **nuScenes** dataset (Green: the smaller the better. Blue: the higher the better).

Conclusion

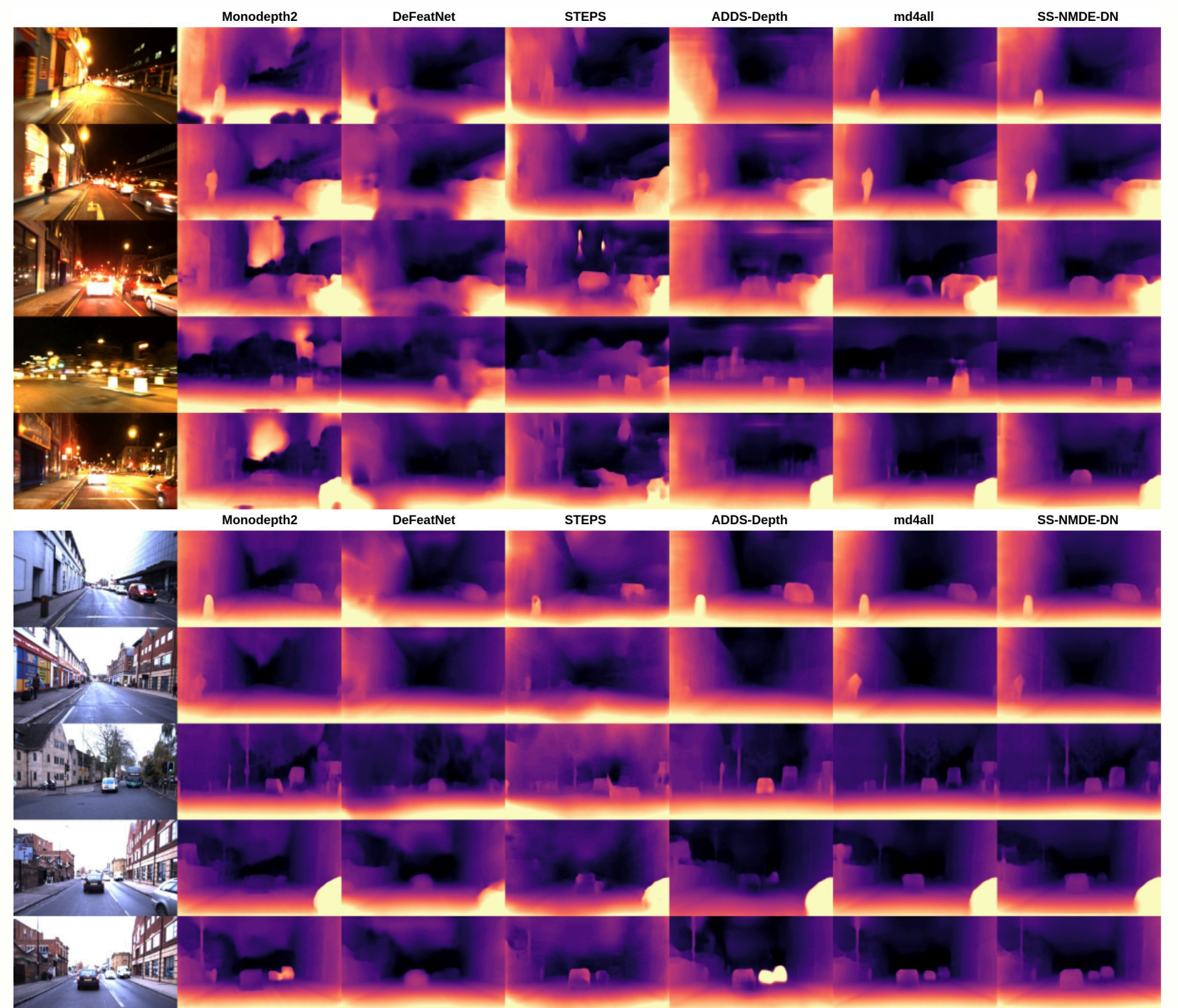
We have proposed a model for day-night (or only night) monocular depth estimation with synthetic images generated by GAN and semantic informations from segmentation network. Experiment results prove our model to be effective and with great development potential.

Method

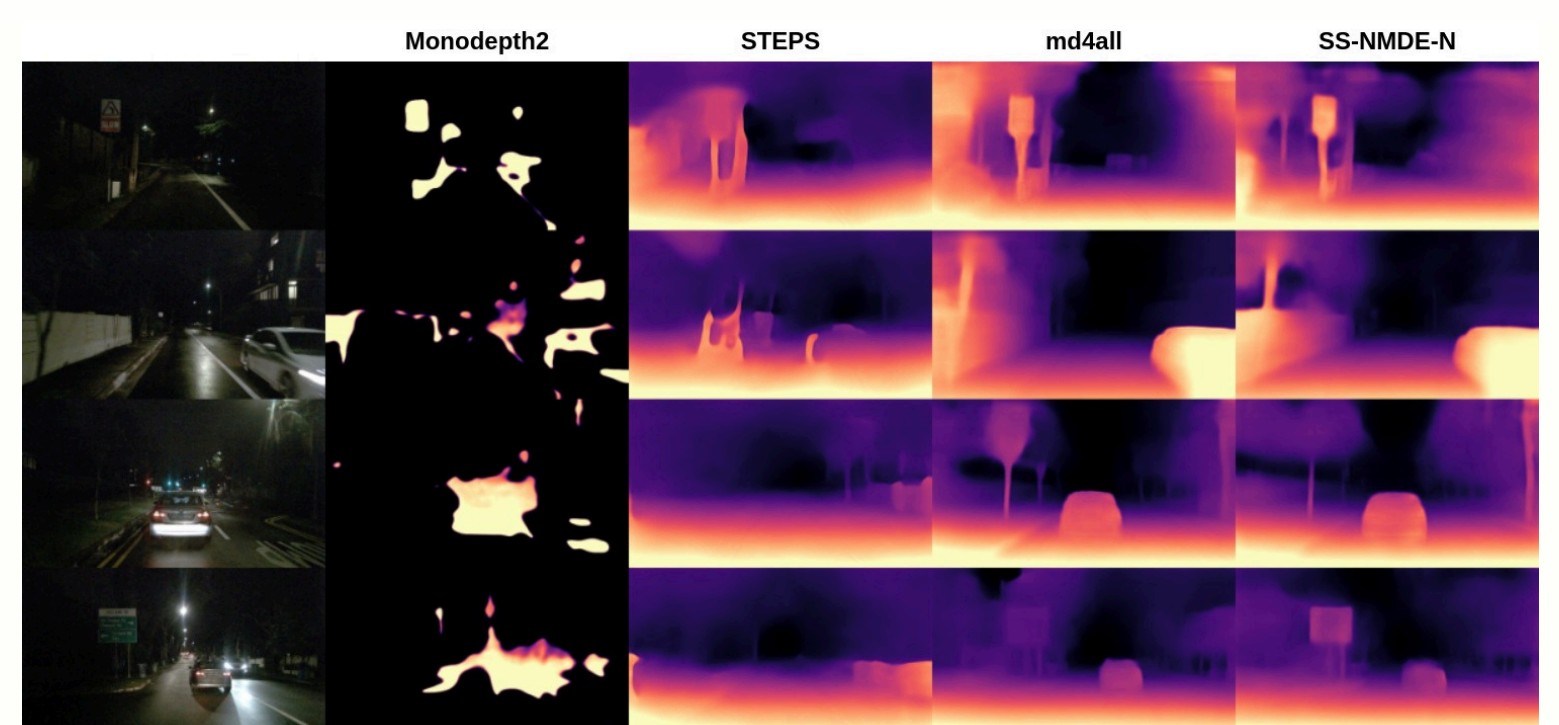
- A Generative Adversarial Network (GAN) generating paired night-time and day-time images.
- A Day-night (or night only) Depth Estimation Network inputting the generated paired images.
- Photometric constraints being applied only in day-time images including camera pose estimation and warping process.
- A pretrained-on-day-time Pseudo Depth network for pseudo supervising the main Depth Network with depth consistency loss.
- A multi-task learning framework with joint depth-segmentation learning.
- A distillation module for boosting depth estimation features.



Visualizations



Visualization results on **Oxford RobotCar** dataset (brighter means closer).



Visualization results on **nuScenes** (brighter means closer).